

A261 – Scale Hierarchy: From ξ to Masses, α , and E_0

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.1 Purpose

[STIPULATION] This document develops the complete scale hierarchy of FFGFT: from the single geometric constant ξ to the characteristic energy scales, to the lepton masses, to the fine-structure constant α , and to E_0 as the logarithmic mean. The core: all these quantities follow from ξ – but the connection is not always achievable by direct cancellation. Ratios (mass ratios, E_0) are correction-free; absolute values need K_{frak} .

.2 The Starting Point: A Single Constant

[B] FFGFT begins with a single dimensionless constant:

$$\xi = \frac{4}{3} \times 10^{-4} = 1.333 \dots \times 10^{-4}.$$

From it all fundamental scales follow in natural units ($\hbar = c = \ell_P = 1$).

[B] Characteristic T0 scales:

$$L_0 = \xi \cdot \ell_P \approx 1.333 \times 10^{-4} \ell_P, \quad E_0^{\text{scale}} = L_0^{-1} = \xi^{-1} \cdot E_P.$$

.3 Lepton Masses from Geometry

[K] The mass formulas (A100) in compact form:

$$m_e = c_e \cdot \xi^{5/2}, \quad m_\mu = c_\mu \cdot \xi^2, \quad m_\tau = c_\tau \cdot \xi^{3/2},$$

where the coefficients have geometric origin:

$$c_e = \frac{3\sqrt{3}}{2\pi\alpha^{1/2}}, \quad c_\mu = \frac{9}{4\pi\alpha}, \quad c_\tau = \frac{27\sqrt{3}}{8\pi\alpha^{3/2}}.$$

[K] Important: apparently simple fractions. In the short form the coefficients appear as rational numbers ($r_e = 4/3$, $r_\mu = 16/5$, $r_\tau = 25/9$ from A100). These fractions are *not* arbitrary and must not be directly cancelled: they encode spatial geometry (π , $\sqrt{3}$) and electromagnetic coupling (α). Both representations (Yukawa short form and extended form) are the same theory in two notations (A130).

.4 E_0 as Logarithmic Mean

[B] The characteristic energy E_0 is the geometric mean of electron and muon mass:

$$E_0 = \sqrt{m_e \cdot m_\mu}.$$

Logarithmically: $\log E_0 = \frac{1}{2}[\log m_e + \log m_\mu]$ – E_0 lies exactly in the middle between m_e and m_μ on a logarithmic scale. Numerically: $E_0 = \sqrt{0.511 \times 105.658} \text{ MeV} \approx 7.35 \text{ MeV}$ (geometric mean). The value from the ξ -path (A130) gives $E_0 \approx 7.398 \text{ MeV}$; this difference is where K_{frak} enters.

[B] From the mass formulas:

$$E_0 = \sqrt{c_e c_\mu} \cdot \xi^{9/4}.$$

E_0 is *correction-free* in the sense: because K_{frak} appears as a common factor in both masses, it does not cancel in the geometric mean – $E_0^{\text{exp}} = K_{\text{frak}} \cdot E_0^{\text{bare}}$, so E_0 itself is an observable that already contains the fractal correction.

.5 The Fine-Structure Constant from ξ

[K] The fundamental relation:

$$\alpha = \xi \cdot E_0^2 = c_e c_\mu \cdot \xi^{11/2}.$$

Substituting the geometric coefficients:

$$\alpha^{5/2} = \frac{27\sqrt{3}}{8\pi^2} \cdot \xi^{11/2}, \quad \text{hence} \quad \alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5}.$$

With fractal correction (A040):

$$\alpha = \left(\frac{27\sqrt{3}}{8\pi^2}\right)^{2/5} \cdot \xi^{11/5} \cdot K_{\text{frak}}, \quad K_{\text{frak}} = 1 - 100\xi \approx 0.9867.$$

Numerically: $\alpha^{-1} \approx 137.036$ – agreement to 0.0005 % (A130).

.6 Why α Does Not Follow Directly from $\xi^{11/2}$ Without Unit Convention

[S] $\alpha \propto \xi^{11/2}$ means: if ξ changed, α would change with exponent $11/2 = 5.5$. That is a strong coupling. The coefficients c_e, c_μ are of order 10^{19} in natural units – that is not a free parameter but the conversion factor between the Planck scale and MeV. In natural units (Planck energies) $c_e \sim 9.67$ and $c_\mu \sim 98.1$ – order 1 to 100, geometrically understandable.

.7 Ratios Are Correction-Free

[B] The mass ratio:

$$\frac{m_\mu}{m_e} = \frac{K_{\text{frak}} \cdot m_\mu^{\text{bare}}}{K_{\text{frak}} \cdot m_e^{\text{bare}}} = \frac{m_\mu^{\text{bare}}}{m_e^{\text{bare}}}.$$

K_{frak} cancels. The same holds for the Koide ratios (A110) and the $g-2$ ratios (A138). This is no coincidence: multiplicative corrections always cancel in ratios.

[K] Absolute values, by contrast, need K_{frak} : without correction $\alpha^{-1} \approx 138.9$ instead of $137.04 - 1.4\%$ deviation. With K_{frak} : exact. Hence K_{frak} is not a post-correction but part of the structure (A040).

.8 The Complete Derivation Chain

[B] All fundamental quantities from ξ in one chain:

$$\xi \xrightarrow{\text{geometry}} r_0, E_0^{\text{scale}} \xrightarrow{\text{quantum numbers}} m_e, m_\mu, m_\tau \xrightarrow{\text{geom. mean}} E_0 \xrightarrow{\xi \cdot E_0^2} \alpha \xrightarrow{\text{K-corr.}} \alpha_{\text{exp.}}$$

Additionally: $\xi \rightarrow L_0 \rightarrow G$ (gravitational constant, A142) and $\xi \rightarrow L_0$ (A015).

.9 Verification Script

- Numerically: $E_0 = \sqrt{m_e m_\mu}$, $\alpha = \xi E_0^2$, ratios correction-free vs. absolute values with K_{frak} :
python/A_Serie_Skripte/a261_skalenhierarchie.py

.10 Open Questions

The coefficients c_e, c_μ, c_τ are not fully independently derived. They contain α explicitly – and α in turn follows from the masses. The circularity is named as a methodological problem in A130; the T0 resolution (common geometric origin) is a claim, not a proof.

The quark coefficients are missing. The c_i structure for quarks is not worked out at the same level as for leptons (A150).

.11 Supersedes

This document subsumes: Dok. 070 (mathematical structure of T0 theory: circular ratios, scale hierarchy from ξ , lepton masses, α derivation, E_0 as logarithmic mean, geometric constant C , why ratios are correction-free). Dok. 093 (β -parameter derivation, self-consistency condition).

.12 Use of AI Tools

This document was created with the assistance of AI tools. Responsibility for content and errors rests with the author. Full wording of the rules in A010.